

MOHAMED AFLI

Embedded Software Engineer — IoT, Firmware & Automotive
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Professional Summary

Embedded Software Engineer with 7+ years of experience developing production firmware for IoT, automotive and real-time systems. Strong C/C++ background on ESP32, STM32 and Nordic nRF52 platforms with FreeRTOS and Zephyr RTOS. Expertise in low-power design, secure OTA updates, battery management systems (BMS), and wireless connectivity (BLE, LoRa, GSM/GPRS, Wi-Fi, MQTT). Trained in AUTOSAR Classic (DaVinci, EB tresos) and ISO 26262 concepts.

Technical Skills

- **Programming:** C, C++, Python, Embedded C, bare-metal and RTOS-based firmware
- **RTOS & Frameworks:** FreeRTOS, Zephyr RTOS, ESP-IDF, Nordic nRF5 SDK, STM32 HAL/LL, STM32CubeMX, LVGL
- **Microcontrollers:** ESP32 (S3, C3), STM32 (F4, F7, WL), Nordic nRF52/53/93M1, ATmega328
- **Protocols & Connectivity:** UART, SPI, I²C, CAN, LIN, SMBus, Modbus, BLE, LoRa, Wi-Fi, GSM/GPRS, TCP/IP, HTTP, MQTT, SSL/TLS
- **Automotive:** AUTOSAR Classic Platform, Vector DaVinci Configurator, EB tresos, ISO 26262
- **Power & BMS:** TI BQ25798, BQ40Z50, bq27z758, MPPT solar charging, State-of-Charge estimation SOC, SOH
- **Tools & Practices:** Git/GitHub, GDB, JTAG, Keil uVision, MATLAB/Simulink, VS Code, OTA update design, Embedded Linux (Yocto)

Professional Experience

Embedded Software Engineer

InspireTech, Sousse, Tunisia

Dec 2025 – Present

HMI & Cloud-Connected Industrial System (ESP32-S3)

- Architected a touchscreen HMI on ESP32-S3 (LVGL) for real-time industrial data acquisition, reducing operator response time by 40%.
- Implemented bidirectional MQTT/HTTP communication with 99.5% uptime and offline data synchronisation covering network outages of 24+ hours.
- Developed a secure OTA update framework, eliminating on-site maintenance visits and cutting deployment costs by an estimated 60%.

Tech: ESP32-S3, ESP-IDF, FreeRTOS, LVGL, MQTT, HTTP, OTA, TCP/IP

Low-Power Wireless Sensor Node (nRF52832)

- Engineered battery-powered sensor-node firmware with 6+ months of autonomy and a custom BLE protocol supporting 50+ devices.
- Implemented on-device sensor filtering (SPI/I²C), reducing cloud payload by 35%; delivered production firmware within an 8-week sprint.

Tech: nRF52832, Nordic SDK, Zephyr RTOS, BLE, SPI, I²C, UART

Garment Steam Finishing Machine (ESP32, ESP32-S3 & ATmega32U4)

- Built a multi-MCU control system for an industrial garment steam finishing machine, coordinating an ESP32 and an ATmega32U4 over RS485.
- Implemented motion control across heterogeneous actuators: NEMA17 and NEMA14 stepper motors driven by TMC2209 drivers, plus BLDC control via L6234 drivers.
- Added homing and safety logic using end switches and IR sensors, with RGB LED status indication for machine states and cycle feedback.

- Developed an HTML web interface on the ESP32-S3 enabling operators to control motors and supervise the machine remotely.

Tech: ESP32, ESP32-S3, ATmega32U4, RS485, TMC2209, L6234, NEMA17, NEMA14, BLDC, IR sensors, end switches, RGB LED, HTML/HTTP, Websocket

Embedded Software Engineer

Jan 2019 – Dec 2025

Tunisian Armed Forces, Tunis, Tunisia

Medical Sterilizer Control, Smart Solar Valve & BMS

2024–2025

- Developed safety-critical medical sterilizer control (ESP32-S3) with LVGL HMI and Firebase/MQTT remote monitoring, improving maintenance response time by 50%.
- Designed a dual-MCU solar irrigation valve (STM32F4 motor control + ESP32-S3 connectivity) achieving 95%+ MPPT efficiency and <2 mA standby current.
- Built two production-ready 4-cell Li-Ion BMS solutions (ESP32 and STM32) with TI BQ25798/BQ40Z50; wrote an SMBus driver reaching $\pm 2\%$ SoC accuracy, validated over 500+ cycles.

Tech: ESP32-S3, STM32F4, ESP-IDF, FreeRTOS, Firebase, Mosquitto, LVGL, BQ25798, BQ40Z50, MPPT, SMBus/I²C

Secure OTA Framework over GPRS

2024

- Architected a modular OTA framework for 100+ field devices over cellular GPRS (ESP32 + Quectel MC60) with reliable 50 KB/s downloads.
- Implemented SSL/TLS with the GitHub Releases API and fault-tolerant flash partitioning with automatic rollback, ensuring no devices were bricked.

Tech: ESP32, ESP-IDF, Quectel MC60, GPRS/HTTP, SSL/TLS, flash memory management

IoT Battery Management Prototype

2023

- Developed an ESP32-C3 BMS prototype with AG105 charger and INA sensors; implemented MQTT telemetry for remote supervision; validated through 200+ hours of testing.

Tech: ESP32-C3, ESP-IDF, AG105, INA sensors, MQTT

Athlete Health Monitoring IoT System

2022

- Built a wearable health tracker (GPS, heart rate, MPU6050, DS18B20) with sensor fusion at <100 ms latency, a GSM/Firebase cloud pipeline and 12+ hours of battery life.

Tech: ESP32, ESP-IDF, FreeRTOS, SIM800C, GPS NEO-6M, MPU6050, DS18B20, Firebase, GSM

Reusable Driver Library & IoT Communication Stack

2020–2021

- Created a reusable ESP-IDF/STM32 driver library that cut new-project development time by 50%; built GSM, LoRa, MQTT and HTTP stacks deployed across 10+ projects.

Tech: ESP-IDF, STM32 (HAL/LL), Keil uVision, MQTT, HTTP, LoRa, GSM

Remote Door Control System

2019

- Designed secure remote access over GSM (SIM800C) and LoRa (RFM) with DC motor control, safety interlocks and limit-switch feedback on ATmega328.

Tech: ATmega328, Microchip Studio, MPLAB X, SIM800C, LoRa RFM, DC motor control

Education

Engineering Degree in Electromechanics Military Academy of Fondouk Jdid, Nabeul, Tunisia Sep 2015 – Mar 2018
Preparatory Classes in Physics & Technology IPEIN, Nabeul, Tunisia 2013 – 2015

Certifications & Training

- AUTOSAR Classic Platform: BSW configuration, DaVinci Configurator, EB tresos
- Embedded Linux & Yocto Project: system customisation, BSP development
- MATLAB & Simulink: model-based design, control-system simulation

Languages

Arabic: Native • English: B2 • French: B2 • German: B1